

BankOS Banking Guide

Introduction

Welcome to BankOS.

BankOS is a digital banking management platform designed to help customers and administrators manage banking operations efficiently. The platform supports customer management, account administration, transfers, debit and credit operations, transaction tracking, and AI-powered assistance.

This guide explains the main banking concepts and application functionalities used inside the BankOS platform.

1. Banking Concepts

1.1 Bank Account

A bank account is a financial record owned by a customer. It stores money and allows financial operations such as deposits, withdrawals, and transfers.

Each bank account contains:

- Account ID
 - Balance
 - Currency
 - Creation date
 - Account status
 - Customer owner
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1.2 Customer

A customer is a person who owns one or multiple bank accounts.

Each customer contains:

- Customer ID
- Full name
- Email address
- Creation date

A customer can own:

- Current accounts
- Saving accounts

2. Account Types

2.1 Current Account

A current account is a standard bank account used for daily operations.

Features:

- Supports deposits and withdrawals
- Supports transfers
- Can have an overdraft limit
- Designed for frequent operations

Example:

A customer with a current account can transfer money, pay bills, and perform debit operations regularly.

2.2 Saving Account

A saving account is designed for saving money over time.

Features:

- Generates interest
- Lower risk
- Used for long-term savings
- Supports credit and debit operations

Example:

If a saving account has an interest rate of 3%, the bank adds interest periodically to the account balance.

3. Banking Operations

3.1 Credit Operation

A credit operation adds money to a bank account.

Examples:

- Salary deposit
- Bank transfer received

- Cash deposit

Effect:

- Account balance increases

Example:

If an account contains 5,000 MAD and receives a credit of 1,000 MAD:

New balance = 6,000 MAD

3.2 Debit Operation

A debit operation removes money from a bank account.

Examples:

- Withdrawal
- Payment
- Transfer sent

Effect:

- Account balance decreases

Example:

If an account contains 5,000 MAD and a debit of 500 MAD is performed:

New balance = 4,500 MAD

3.3 Transfer Operation

A transfer operation moves money from one account to another.

A transfer contains:

- Source account
- Destination account
- Amount
- Description
- Date

Steps:

1. Debit source account
2. Credit destination account

3. Save transaction history

4. Account Status

BankOS supports multiple account statuses.

CREATED

The account has been created but not yet activated.

ACTIVATED

The account is active and can perform operations.

SUSPENDED

The account is temporarily disabled.

Suspended accounts cannot:

- Perform transfers
- Perform debit operations
- Receive some banking services

5. Security Rules

BankOS uses JWT authentication to secure the platform.

Features:

- Secure login
- Token-based authentication
- Role-based authorization
- Protected REST APIs

Users must authenticate before accessing banking operations.

6. Dashboard Analytics

The dashboard displays:

- Number of customers
- Number of accounts

- Total balance
- Number of operations
- Credit statistics
- Debit statistics
- Active accounts
- Suspended accounts

Charts are generated using Chart.js.

7. Artificial Intelligence Assistant

BankOS integrates an AI assistant based on Spring AI and OpenAI.

The AI assistant can:

- Explain banking concepts
- Answer banking questions
- Help users navigate the platform
- Retrieve banking information using tools
- Use RAG (Retrieval Augmented Generation)

The assistant uses:

- OpenAI GPT models
 - Vector store embeddings
 - PDF document indexing
 - Spring AI tools
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8. RAG (Retrieval Augmented Generation)

RAG allows the AI assistant to answer questions using banking documentation.

Workflow:

1. Documents are loaded
2. Documents are split into chunks
3. Embeddings are generated
4. Chunks are stored in a vector store
5. User questions are matched against the vector store
6. Relevant context is sent to the AI model

Benefits:

- More accurate answers
 - Context-aware responses
 - Banking-specific knowledge
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9. Frequently Asked Questions

What is the difference between a current account and a saving account?

A current account is used for daily operations and may support overdraft.

A saving account is designed for storing money and generating interest.

What happens during a debit operation?

Money is removed from the account balance.

What happens during a credit operation?

Money is added to the account balance.

Can a customer own multiple accounts?

Yes. A customer can own several current and saving accounts.

Why can an account be suspended?

Accounts may be suspended for:

- Security reasons
 - Suspicious activity
 - Administrative decisions
 - Policy violations
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10. API Examples

Get All Customers

GET /customers

Get Customer By ID

GET /customers/{id}

Get Account By ID

GET /accounts/{id}

Perform Debit Operation

POST /accounts/debit

Perform Credit Operation

POST /accounts/credit

Perform Transfer

POST /accounts/transfer

11. Technologies Used

Backend

- Spring Boot 3
- Spring Security
- Spring Data JPA
- Spring AI
- JWT
- MySQL

Frontend

- Angular 17
- TypeScript
- RxJS
- Chart.js

AI Stack

- OpenAI API
 - Spring AI Agent
 - SimpleVectorStore
 - PDF Document Reader
 - RAG
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12. Conclusion

BankOS is an educational banking management platform combining modern web technologies, secure authentication, banking operations, analytics dashboards, and artificial intelligence.

The project demonstrates:

- Full-stack web development
- REST API architecture
- JWT security
- Database management
- AI integration using Spring AI
- RAG-based intelligent assistance